

[Aim: 100/100 in Maths]

आरंभ

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Lecture #9
[Miscellaneous Questions]

#LP: If $ax + by = a^2 - b^2$ and $bx + ay = 0$, then the value of $x + y$ is:

(i) $a^2 - b^2$

(ii) $a + b$

(iii) $a - b$

(iv) $a^2 + b^2$

⊕

$$ax + by = a^2 - b^2$$

⊕

$$bx + ay = 0$$

$$ax + bx + by + ay = a^2 - b^2$$

$$x(a+b) + y(a+b) = (a+b)(a-b)$$

$$(a+b)(x+y) = (a+b)(a-b)$$

$$x + y = a - b \quad \text{--- (I)}$$

$$ax + by = a^2 - b^2$$

$$bx + ay = 0$$

$$ax - bx + by - ay = a^2 - b^2$$

$$x(a-b) + y(b-a) = a^2 - b^2$$

$$x(a-b) - y(a-b) = (a+b)(a-b)$$

$$(a-b)(x-y) = (a+b)(a-b)$$

$$x - y = a + b \quad \text{--- (II)}$$

$$\begin{matrix} x = \\ y = \end{matrix}$$

Final Ans $\Rightarrow x + y$

#LP: Ankita travels 14 km to her home partly by rickshaw and partly by bus. She takes half an hour if she travels 2 km by rickshaw and the remaining distance by bus. On the other hand, if she travels 4 km by rickshaw and the remaining distance by bus, she takes 9 minutes longer. Find the speed of the rickshaw and of the bus.

Case I

Total dist. = 14 km

Let $S_r = x$ km/hr & $S_b = y$ km/hr

Total time = $\frac{1}{2}$ hr

$$t(2 \text{ km by rickshaw}) + t(12 \text{ km by bus}) = \frac{1}{2} \text{ hr}$$

$$\frac{2 \text{ km}}{x \text{ km/hr}} + \frac{12 \text{ km}}{y \text{ km/hr}} = \frac{1}{2} \text{ hr}$$

$$\boxed{\frac{2}{x} + \frac{12}{y} = \frac{1}{2}} \quad \text{--- (1)}$$

$\frac{9}{60} \text{ hr}$

Case II

$$t(4 \text{ km by rickshaw}) + t(10 \text{ km by bus}) = \frac{1}{2} + \frac{9}{60}$$

$$\boxed{\frac{4}{x} + \frac{10}{y} = \frac{1}{2} + \frac{9}{60}}$$

$$\frac{4}{x} + \frac{10}{y} = \frac{30 + 9}{60}$$

$$\boxed{\frac{4}{x} + \frac{10}{y} = \frac{39}{20}} \quad \text{--- (2)}$$

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$$\frac{2}{x} + \frac{12}{y} = \frac{1}{2}$$

$$2 \times \frac{1}{x} + 12 \times \frac{1}{y} = \frac{1}{2}$$

$$2p + 12q = \frac{1}{2}$$

$$\frac{1}{x} = p$$
$$\frac{1}{y} = q$$

$$\frac{4}{x} + \frac{10}{y} = \frac{13}{20}$$

$$4 \times \frac{1}{x} + 10 \times \frac{1}{y} = \frac{13}{20}$$

$$4p + 10q = \frac{13}{20}$$

$$\frac{1}{x} = p = \frac{1}{10}$$

$$\frac{1}{y} = q = \frac{1}{40}$$

$$S_{bus} \rightarrow y = 40 \text{ km/hr}$$

$$x = 10 \text{ km/hr} \leftarrow S_{rich}$$

$$d = s \times t$$

#LP: A train covered a certain distance at a uniform speed. If the train had been 6 km/hr faster, it would have taken 4 hours less than the scheduled time and if the train were slower by 6 km/hr, it would have taken 6 hours more than the scheduled time. Find the length of the journey. (CBSE 2020)

Original speed = x km/hr $\xrightarrow{\text{I}}$ $(x+6)$ km/hr
 original time = y hr $\longrightarrow$ $(y-4)$ hr

$$d = s \times t$$

$$d = xy$$

original distance will remain same

$$d = (x+6)(y-4)$$

$$xy = xy - 4x + 6y - 24$$

$$4x - 6y = -24$$

$$2(2x - 3y) = -12$$

$$2x - 3y = -12 \quad \text{--- (2)}$$

II $(x-6)$ km/hr
 $(y+6)$ hr

$$d = (x-6)(y+6)$$

$$xy = xy + 6x - 6y - 36$$

$$36 = 6x - 6y$$

$$6x = 6(x-y)$$

$$x - y = 6 \quad \text{--- (1)}$$

Final Ans = $d = xy =$

#LP: Vijay had some bananas and he divided them into two lots A and B. He sold the first lot at the rate of Rs. 2 for 3 bananas and the second lot at the rate of Rs 1 per banana and got a total of Rs 400.

~~If he had sold the first lot at the rate of Rs 1 per banana and the second lot at the rate of Rs 4 for 5 bananas, his total collection would have been Rs 400, Find the total number of bananas he had.~~

Lot A $\Rightarrow$ x bananas
Lot B $\Rightarrow$ y bananas $\rightarrow$ total $= (x + y)$

Case I

Total collection = 400/-

Money of A + Money of B = 400

$$\boxed{\frac{2}{3}x + y = 400 \text{ --- (I)}}$$

Lot A

3 bananas $\Rightarrow$ ₹ 2

1 ban $\Rightarrow$ ₹ $\frac{2}{3}$

x ban $\Rightarrow$ ₹ $\left(x \times \frac{2}{3}\right)$

Lot B

1 ban $\Rightarrow$ ₹ 1

y ban $\Rightarrow$ ₹ $(y \times 1)$

#LP: Vijay had some bananas and he divided them into two lots A and B. He sold the first lot at the rate of Rs 2 for 3 bananas and the second lot at the rate of Rs 4 for 5 bananas and got a total of Rs 460. If he had sold the first lot at the rate of Rs 1 per banana and the second lot at the rate of Rs 4 for 5 bananas, his total collection would have been Rs 460, Find the total number of bananas he had.

Lot A $\Rightarrow$ x bananas
 Lot B $\Rightarrow$ y bananas
 $\rightarrow$ total = $(x + y)$

Lot A

1 ban = ₹1

x ban = ₹ $(x \times 1)$

Case II

Total = 460

$$x + \frac{4}{5}y = 460 \quad \text{--- (II)}$$

Lot B

5 ban $\rightarrow$ ₹4

1 ban $\rightarrow$ ₹ $\frac{4}{5}$

y ban $\rightarrow$ ₹ $\left(y \times \frac{4}{5}\right)$

#LP: Priya was attempting a test consisting of only 'Multiple Choice Questions'. One mark is awarded for every correct answer, while $\frac{1}{4}$ mark is deducted for every wrong answer. Priya knew the answers to some of the questions. The rest she attempted by guessing. She answered 100 questions in total.

(i) Priya knew the answers to 60 questions, and if half of the answers she attempted by guessing were correct, then how many marks did she get?

Correct $\Rightarrow +1$ marks
Wrong $\Rightarrow -\frac{1}{4}$ marks
Total Q $\Rightarrow 100$ questions

i) Total Q $\Rightarrow 100$

Correct $\checkmark 60$

40 guess

20 C 20 in.

Total correct $\rightarrow 60 + 20 = 80$

Total incorrect $\rightarrow 20$

Total Marks $\Rightarrow 80 \times 1 - \frac{1}{4} \times 20 = 75$ marks

(ii) If all the questions she attempted by guessing were wrong and she answered 60 correctly, then how many marks did she get?



$$\text{Total Marks} \Rightarrow 60 \times 1 - \frac{1}{4} \times 40$$
$$\Rightarrow \underline{50 \text{ marks}}$$

(iii) If all the questions she attempted by guessing were wrong, then how many questions did she answer correctly to score 65 marks?

Total $\Rightarrow 100$ Q
let correct $\Rightarrow x$ Q
Incorrect $\Rightarrow (100 - x)$

$$\text{Marks} = 65$$
$$\Rightarrow x \times 1 - \frac{1}{4}(100 - x) = 65$$
$$\Rightarrow x - \frac{1}{4} \times 100 + \frac{1}{4}x = 65$$
$$\Rightarrow x + \frac{x}{4} = 65 + 25$$

$$\frac{5x}{4} = 90$$
$$x = 72 \text{ Q correct}$$

#LP: The incomes of two persons are in the ratio 11:9. Their expenditures are in the ratio 5:4. Each saves Rs. 300 per month. Find their monthly incomes.

$$\frac{I_A}{I_B} = \frac{11}{9}$$

$$\frac{E_A}{E_B} = \frac{5}{4}$$

let $I_A = 11x = 11x$ ✓
 $I_B = 9x = 9x$ ✓

let $E_A = 5y$
 $E_B = 4y$

$$(\text{Saving})_A = 300$$

$$I_A - E_A = 300$$

$$11x - 5y = 300 \quad \text{--- (2)}$$

$$(\text{Saving})_B = 300$$

$$I_B - E_B = 300$$

$$9x - 4y = 300 \quad \text{--- (1)}$$

Final Ans

$x =$

PLEASE **रहा** !

THANK YOU COODIESS !!

